Carrier-Weighted Networks: Separating Coherent Minority Structure from Incoherent Edges with Matched Pooled Correlations

Yong-Wook Shin
Department of Psychiatry, Asan Medical Centre,
University of Ulsan College of Medicine, Seoul, Korea

August 28, 2026

Abstract

In networks built from questionnaire data, a pooled edge correlation — the correlation of one item pair — is an average over respondents: the same value can arise from weak coupling in everyone, strong coupling in a coherent minority, or accidental co-endorsement by scattered, mutually unrelated subsets. Pooled network pipelines cannot tell these apart. We introduce an edge-level estimator of carrier coherence that requires no prespecified respondent groups, $w_e^* = |\rho_e|(1 - G_e)\max(\bar{O}_e, 0)$: the product of the pooled correlation magnitude $|\rho_e|$, an evenness-of-support factor $(1-G_e)$, and a coherence factor $\max(\bar{O}_e, 0)$ measuring whether the respondents who carry an edge also carry the neighbouring edges that share an item with it. In a preregistered simulation with pooled correlations matched by calibrated design, the estimator separated minority-carried community edges from planted incoherent edges ("distractors") with an area under the receiver operating characteristic curve (AUROC) of .985 versus .469 for raw correlations (paired difference $+.516$, 95% confidence interval $[+.506, +.526]$). A concentration-only variant of the estimator remained at chance while a coherence-only variant matched the full estimator — the separation is due to the coherence factor alone. Community recovery by the full pipeline showed no improvement at $n = 600$: by design, both pipelines drew their candidate edges from the same raw-correlation filter, so reweighting could not restore true edges that the selection step had already discarded.

1 Introduction

A correlation between two questionnaire items is an average over respondents. The estimate $|\rho| = .45$ is compatible with at least three different data-generating stories: (a) a weak coupling present in essentially every respondent; (b) a strong coupling present in a coherent 30% minority and absent elsewhere; and (c) no population-wide coupling at all, but incidental co-endorsement by scattered subsets of respondents who have nothing else in common. Standard psychometric network pipelines — regularized partial correlations or a maximally filtered graph, followed by community detection (Epskamp and Fried, 2018; Massara et al., 2017; Pons and Latapy, 2006; Golino and Epskamp, 2017) — operate entirely on the pooled value and are therefore blind to the distinction. This is an edge-level instance of the familiar warning that group-level statistics need not describe any individual: the ecological fallacy in its psychometric form (Molenaar, 2004; Fisher et al., 2018).

The blindness has two concrete costs. First, communities carried by a coherent minority are attenuated at the pooling stage and are absorbed into neighbouring communities or removed when the network is pruned to its strongest edges (sparsification), because each of their edges looks individually weak. Second, edges of type (c) — pooled correlations without any common set of respondents supplying them; hereafter *distractor* edges — pass every pooled screen and enter the network as if they were structure. Both errors leave no trace in pooled fit statistics: the network that results is internally consistent and wrong.

This paper contributes an edge-level estimator of carrier coherence that requires no pre-specified respondent groups. The estimator separates coherent minority-carried structure from incoherent edges whose pooled correlations match, and it can be incorporated into the standard bootstrap exploratory graph analysis (bootEGA) pipeline by replacing the original edge weights (Christensen and Golino, 2021). Unlike mixture or latent-class network models, it neither partitions respondents nor requires a prespecified group-membership variable; the carriers of different edges may overlap, nest, or cross-cut freely. It is also distinct from group-comparison methods (Jamison et al., 2024) and meta-analytic techniques: the heterogeneity of interest is between respondents within a single sample. The estimator changes only the weights of edges retained by the pipeline's candidate-selection step, leaving community detection and resampling-based inference to the downstream pipeline.

2 The carrier-weighted estimator

2.1 Setup and carrier profiles

Let $x_{rj} \in \{0, 1\}$ be the response of respondent r to item j, and let ρ_e be the tetrachoric correlation for the item pair $e = (j, k)$. Define the per-respondent contribution of respondent r to edge e as

$$S_{re} \;=\; (x_{rj} - \bar{x}_j)\,(x_{rk} - \bar{x}_k), \tag{1}$$

the r-th summand of the covariance. The vector $S_{\cdot e} = (S_{1e}, \ldots, S_{ne})$ is the *carrier profile* of the edge: it records who supplies the association. Because the summand is centred, the profile is a signed contribution vector, not an endorsement indicator: a respondent endorsing both items and a respondent endorsing neither both contribute positively (the $(0, 0)$ cell contributes $\bar{x}_j\bar{x}_k$). "Carrier" is a mnemonic for the respondents whose contributions dominate the positive mass of this vector; every quantity below is computed from the full signed profile, never from a thresholded carrier set. The estimator is deliberately hybrid in scale: the magnitude factor uses the tetrachoric ρ_e (latent scale), while the profile is computed on the observed binary responses, where the respondent dimension still exists. Pooling collapses this vector to its mean; everything the present method does is done before that collapse.

2.2 Two carrier components

Two properties of the carrier profile carry the diagnostic information.

Concentration. The Gini coefficient G_e of the magnitudes $|S_{re}|$[1] measures how unevenly the edge's support is distributed over respondents. An edge carried by everyone has low G_e; an edge carried by a small subset — coherent or not — has high G_e. The even-support factor $(1 - G_e)$ down-weights concentrated edges.

[1]Computed in the standard sorted-index form: with $v_{(1)} \leq \cdots \leq v_{(n)}$ the sorted magnitudes $|S_{re}|$, $G_e = 2\sum_r r\,v_{(r)}/\big(n\sum_r v_{(r)}\big) - (n+1)/n$, with no small-sample correction.

Coherence. Let $\mathcal{N}(e)$ be the set of edges that share a node with e — the neighbours of e in the line graph of the network (the derived graph whose nodes are the original network's edges). $\bar{O}_e$ is the mean cosine between the carrier profile of e and those of its neighbours:

$$\bar{O}_e \;=\; \frac{1}{|\mathcal{N}(e)|} \sum_{f \in \mathcal{N}(e)} \frac{\langle S_{\cdot e},\, S_{\cdot f} \rangle}{\|S_{\cdot e}\| \, \|S_{\cdot f}\|}. \tag{2}$$

Only node-sharing pairs enter the average — for $e = (j,k)$ that is $\deg(j) + \deg(k) - 2$ cosines, rather than a comparison of e with all $|E| - 1$ other edges — so the statistic is local and computationally inexpensive even in dense networks; the structural reason for this restriction is given in Section 2.3. $\bar{O}_e$ asks the only question that concentration cannot: *do the same respondents carry the neighbouring edges?*

2.3 Why neighbour alignment: the wedge — a node-sharing edge pair — is the minimal detection unit

A single edge, considered alone, contains no information that discriminates scenario (b) from scenario (c) of the Introduction. The pooled ρ_e has already collapsed the respondent dimension. Concentration does not help either: a real minority-carried edge and an accidental co-endorsement are *both* concentrated, so G_e assigns them similar values. This is the a-priori reason to expect a concentration-only weighting to fail — a prediction the ablation in Section 3.4 tests directly.

The discriminating evidence lives one level up. Latent factors load on *nodes* (items), not on edges. If a factor F loads on item j, it leaves its trace simultaneously on every edge through j — (i,j), (j,k) — and, crucially, that trace passes through the *same* respondents, namely those in whom F is active. The shared node is the conduit of the common cause. A distractor edge's carriers, by contrast, have no structural reason to overlap with the carriers of any adjacent edge, so its expected alignment is zero.

The minimal unit of factor detection is therefore not the edge but the node-sharing edge pair — the *wedge*. Just as a correlation is a two-node statistic, carrier coherence is a two-edge statistic, one order up: $\bar{O}$ is a statistic on the line graph of the network, and the estimator's essence is to lift inference from node pairs to edge pairs. A community, in this view, is a connected set of high-coherence wedges sharing one latent cause, and carrier alignment is the empirical fingerprint of that sharing.

2.4 The full weight

The estimator, locked in this form by the preregistration (Section 3), combines the pooled magnitude with both carrier components:

$$w_e^* \;=\; |\rho_e| \, (1 - G_e) \, \max(\bar{O}_e,\, 0). \tag{3}$$

The positive-part rule — a gate that removes edges with $\bar{O}_e < 0$ — reflects a requirement of the downstream community-detection algorithm: walktrap (Pons and Latapy, 2006) interprets a positive weight as attraction, so an edge whose carriers systematically *oppose* those of its neighbours is removed rather than passed through as a small positive weight.

2.5 The minimal toy example: one triangle, two stories

If the wedge is the minimal detection unit, the natural worked example is the smallest structure made entirely of wedges: the closed triangle on three items i, j, k, whose three edges (i, j), (j, k), (i, k) form three node-sharing pairs, so that every edge's $\bar{O}$ is the mean of its two wedge cosines. Closing the triangle also closes a loophole a two-edge example leaves open: with an open wedge, the unused third pair (i, k) can still differ between scenarios, so a sceptic could ask whether triadic closure alone — the presence or absence of the third edge completing the triangle — would have separated them. Here it cannot, by construction. Figure 1 shows two deterministic scenarios on $n = 18$ respondents (no random seed) in which *every* pooled 2×2 contingency table — all three edges, both scenarios — has the same four cells. All pooled pairwise quantities are therefore identical as a matter of arithmetic identity, not approximate matching: every ϕ coefficient is .25, every tetrachoric correlation .40, every Gini coefficient .29, and all item marginals coincide. Every S_{re} entry is a multiple of 1/9; the whole example can be verified by hand in a few minutes.

In scenario A one minority (respondents 1–3) endorses all three items and carries all three edges, while nine singleton respondents (three per item) endorse one item each, filling the off-diagonal cells. All three carrier cosines equal $+2/3$, so $\bar{O} = +2/3$ and every edge survives with $w^* = .19$. In scenario B the same pooled tables are produced by three *disjoint* minorities — respondents 1–3 co-endorse (i, j), 4–6 (j, k), 7–9 (i, k): every pairwise carrier cosine equals $-1/3$, so $\bar{O} = -1/3$ and all three edges are removed. No pooled pairwise quantity — correlation, concentration, marginal, or any function of them, including triadic closure — distinguishes the panels; the eighteen respondents do. (The full three-way $2 \times 2 \times 2$ table does differ, as it must: A has three respondents in the $(1, 1, 1)$ cell, B has none. The estimator recovers exactly this respondent-level information from pairwise objects, without ever forming the higher-order table.) Note that scenario B's edges are perfectly real *within their own carriers* — what the carrier profiles withhold is evidence of a *shared* cause, and that is exactly the evidence the estimator weights. This point returns at scale as the ground-truth (oracle) diagnostic of Section 3.4.

The estimator does not address cancellation across opposite-signed carrier groups or within-person dynamics (Molenaar, 2004); these limitations are discussed in Section 5.

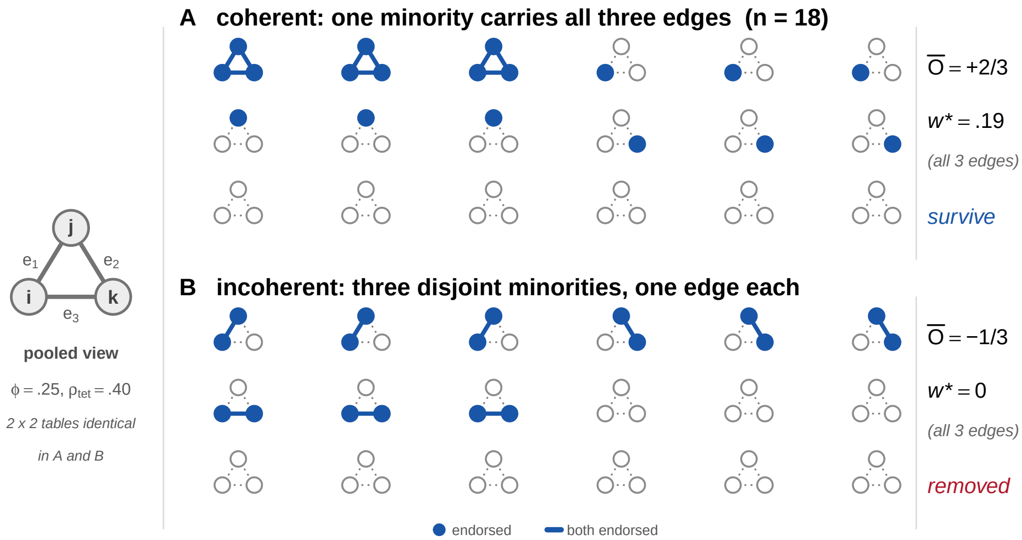

Figure 1: The minimal toy example: one closed triangle on items i, j, k, $n = 18$, two deterministic scenarios whose pooled 2×2 tables are *identical* on every edge (Section 2.5 gives the shared values). Each respondent is drawn as their own copy of the triangle, its outline dotted: filled nodes mark endorsed items; an edge is drawn solid if and only if the respondent endorses both of its endpoints (the respondent *carries* that edge); the pooled view (left, drawn once) is identical for the two scenarios. **A**: one minority carries all three edges, all carrier cosines are $+2/3$, and every edge survives ($w^* = .19$). **B**: three disjoint minorities carry one edge each, all carrier cosines are $-1/3$, and the gate removes every edge. No pooled pairwise statistic distinguishes the panels; the carrier profiles do.

3 Preregistered simulation

The simulation was preregistered before implementation, with the estimator, data-generating model, methods, outcomes, and decision rules locked in advance; one amendment, made before any confirmatory replicate was run, is described below and in Appendix A. The scientific question is the one the toy example poses: can the locked estimator distinguish a real community whose edges are carried by the same minority of respondents from equally strong pooled edges carried by mutually incoherent subsets?

3.1 Design

Fifteen binary items form three equal blocks. Blocks A and B are population-wide: for $j \in A$, $z_{rj} = \lambda F_{r,A} + \epsilon_{rj}$ with $\lambda = .75$ and $F_{r,\cdot}$, ϵ_{rj} independent standard normal (standardized loading $\lambda^* = \lambda/\sqrt{\lambda^2+1} = .60$), and analogously for B. Block C is minority-carried: a single respondent indicator $M_r \sim \text{Bernoulli}(\pi_C)$, $\pi_C = .30$, governs every C item, $z_{rj} = \lambda_C M_r F_{r,C} + \epsilon_{rj}$ with $\lambda_C = 1.10$ (standardized loading .74 *within* carriers; the unstandardized $\lambda_C > \lambda$ compensates the π_C dilution, and the pooled within-C correlation nonetheless remains below the within-A/B level), so that all ten within-C edges — the edges among the five block-C items, items 11–15 — share one carrier subgroup. Six distractor pairs, fixed in advance and spanning blocks, each receive a pair-specific factor active only in its own disjoint respondent subset: for both items j of pair d, $z_{rj} \leftarrow z_{rj} + \delta\, Q_{r,d}\, D_{r,d}$, where $Q_{r,d}$ indicates membership in pair d's subset (prevalence π_D) and $D_{r,d}$ is the pair factor — matched pooled correlation with *no* shared carriers. Latent responses are dichotomized at fixed thresholds: within each block, one archived random arrangement of $(-.35, -.35, 0, .35, .35)$ across its five items, drawn once and then frozen. The

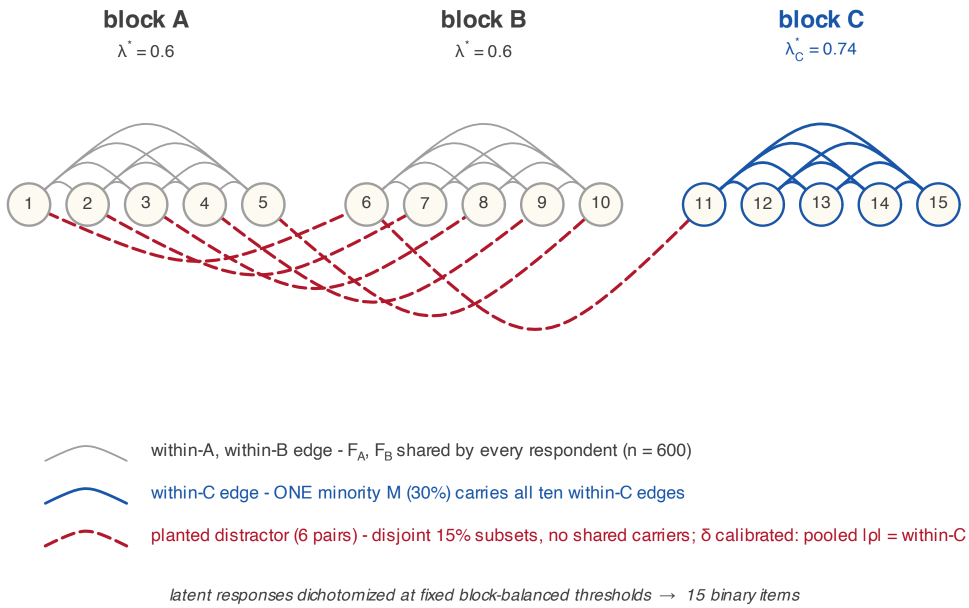

Figure 2: Generative design of the preregistered simulation. Fifteen binary items in three blocks of five; within-block edges are drawn as arcs above the item row, the six planted distractor pairs $(1,6),(2,7),(3,8),(4,9),(5,10),(6,11)$ as dashed arcs below. Blocks A and B are population-wide: every respondent ($n = 600$) shares F_A, F_B (standardized loading $\lambda^* = .60$). Block C is minority-carried: one subgroup M ($\pi_C = .30$) carries all ten within-C edges ($\lambda^*_C = .74$ within carriers). Each distractor pair receives a pair-specific factor active only in its own disjoint 15% respondent subset — no shared carriers — with loading δ calibrated in a discarded pilot so that mean pooled $|\rho|$ matches the within-C mean. The design scales the toy contrast of Figure 1: block C realizes the coherent-minority scenario of panel A, the distractors realize the disjoint-minorities scenario of panel B, and pooled correlation cannot separate them.

distractor loading δ is calibrated in a discarded pilot so that the mean pooled $|\rho|$ of the distractors matches the within-C mean, then frozen. Figure 2 summarises the design.

One amendment was required at the pilot stage. The preregistered zero-mean pair factor $D \sim N(0,1)$ saturates under dichotomization: only half of a distractor's subset can ever reach the concordant-endorsement cell, capping the achievable pooled correlation below the calibration target for every δ. The amendment — made before any confirmatory replicate, under the preregistration's own revision rule — shifts the pair factor to $D \sim N(1,1)$ (an incoherent minority that jointly *endorses* both items, which is the scientific intent of the distractor) and raises π_D from .10 to .15, the prevalence at which the target becomes reachable. With these values, $\delta = 3.75$ achieved calibration to within .001 and was frozen. The confirmatory run used $n = 600$ and 1,000 replicates, with a 1,000-replicate no-structure control, all under preregistered date-stamped seeds (see the Reproducibility statement); no replicate failed.

3.2 Two evaluation levels

The production pipeline first selects candidate edges with the Triangulated Maximally Filtered Graph (TMFG) on raw $|\rho|$ (Massara et al., 2017) — a standard filtering step that retains a planar, well-connected subset of the strongest edges and discards the rest — and only then reweights retained edges; carrier weighting cannot rescue an edge TMFG has already removed. The preregistration therefore mandates two evaluations that may never be collapsed: an *estimator-level* test on a fixed candidate set containing all true within-block and distractor edges, and an *end-to-end* test (raw-$|\rho|$ TMFG $\rightarrow$ reweight $\rightarrow$ walktrap) with TMFG availability — the proportion of the ten true within-C edges retained — reported separately. One structural fact sharpens the second level: the ten within-C edges form the complete graph on the five block-C items (K_5), which cannot be drawn in the plane without crossings; since TMFG returns only planar graphs, it can retain at most nine of the ten within-C edges. Availability is capped at .9 by construction, and the question is how far below that cap the raw-correlation filter falls.

3.3 Methods and outcomes

Four methods — each usable in practice, since none requires knowing the true carriers — receive identical data: raw $|\rho_e|$; the concentration-only ablation $|\rho_e|(1 - G_e)$; the coherence-only ablation $|\rho_e| \max(\bar{O}_e, 0)$ (an ablation removes one factor of the estimator, to locate which factor does the work); and the full carrier-weighted (CW) estimator, Eq. (3). The ablations are mandatory because $(1 - G)$ rewards diffuse support and could in principle penalize the very minority concentration the method aims to recover. A fifth comparator is an oracle — it is handed the true carrier subgroup, information no real analysis could have — and computes correlations within that subgroup, serving as a diagnostic only. Estimator-level outcomes are the area under the receiver operating characteristic curve (AUROC) for ranking the ten within-C edges above the six distractors, the mean score separation, and the mechanism check (distributions of $|\rho|$, G, and $\bar{O}$ from Eq. (2) for within-C versus distractor edges); end-to-end outcomes are the adjusted Rand index (ARI; a chance-corrected agreement between the recovered and true item groupings) against the true partition, recovery of the correct number of communities (k=3), exact recovery of block C, and TMFG availability. All comparisons are paired by replicate with percentile-bootstrap confidence intervals (CI; 10,000 resamples).

The preregistration condenses these outcomes into four decision rules, locked before any confirmatory replicate together with the interpretation of every pass/fail pattern. Rule 1 is the estimator-level claim: CW must rank within-C edges above distractors better than raw correlation (AUROC CW > raw). Rule 2 is the end-to-end claim: CW must improve *both* ARI and exact-C recovery over the raw pipeline. Rule 3 is the specificity control: in no-structure data, CW must not inflate false k=3 detections by more than .05 over raw. Rule 4 is the mechanism check: pooled $|\rho|$ must be matched between within-C edges and distractors (the calibration held), and the separation, if any, must be carried by $\bar{O}$ rather than by G. Results are reported below rule by rule.

3.4 Results

Rule 1 passed: AUROC .985 for CW versus .469 for raw, a paired-bootstrap difference of +.516 with 95% CI $[+.506, +.526]$; the companion outcome, mean score separation (mean weight of within-C edges minus distractors), was +.013 for CW versus $-.007$ for raw. Rule 2 *failed*: the ARI difference +.0034 $[+.0001, +.0067]$ excludes zero, but the exact-C difference +.0020 $[-.0080, +.0120]$ does not. Rule 3 passed: false k=3 rates were .295 (raw) and .308 (CW), a difference of +.013. The absolute rates warrant a comment the rule itself does not require: with the C loadings set to zero, the six distractor pairs remain in the data, and the candidate-selection and community-detection stages shared by both pipelines assemble a spurious third community in roughly 30% of null replicates. That rate is a property of TMFG-plus-walktrap on null data, essentially unchanged by reweighting; the rule targets the increment, which is the only quantity the estimator could affect. Rule 4 passed: pooled $|\rho|$ was matched (C .169, distractor .176), Gini did *not* separate the classes (.172 / .165), and $\bar{O}$ did (.102 / $-$.003).

Rule 4 is the wedge argument of Section 2.3 confirmed at scale (Figure 3): with pooled magnitude matched by design, concentration is indistinguishable between real minority edges and distractors — the concentration-only ablation achieves AUROC .464, chance — while coherence separates them cleanly, and coherence-only (.985) matches the full estimator (Figure 4). The oracle comparator returned an *inverted* AUROC of .002: within-carrier correlation is higher for distractor cliques than for within-C edges at the calibrated δ, because both mechanisms are perfectly real within their own carriers. Only cross-edge coherence distinguishes them — which is the thesis, so the oracle is reported as a diagnostic, not a competitor.

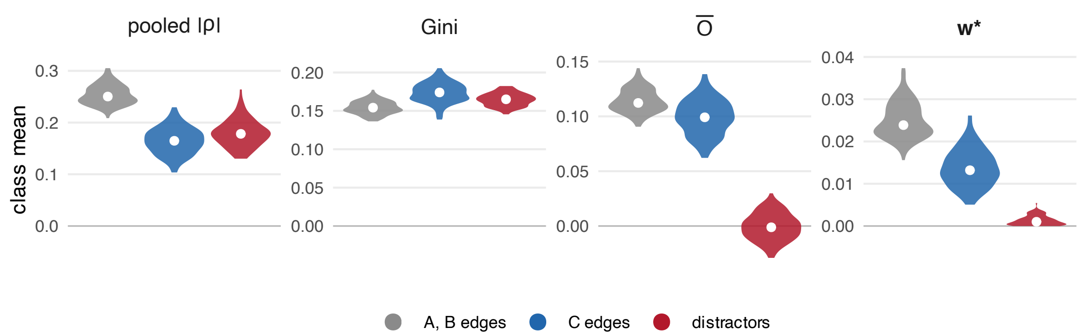

Figure 3: Mechanism check, all 36 structural edge classes (per-replicate class means over the 100 archived per-edge replicates; C and distractor values agree with the 1,000-replicate summary statistics). Pooled $|\rho|$ is matched between minority-carried within-C edges and incoherent distractors by calibrated design, with both below the population-wide A/B level; Gini concentration separates none of the three classes; the coherence statistic $\bar{O}$ alone isolates the distractors, while population-wide A/B edges and minority-carried within-C edges are both coherent. The resulting weight w^* (rightmost panel) preserves the A/B and within-C edges and drives the distractors to near zero: survival is decided by the coherence gate, not by magnitude or concentration. Class separability is quantified by the edge-level AUROC values reported in the text, not by significance tests on replicate means, whose p-values would reflect only the number of simulation replicates.

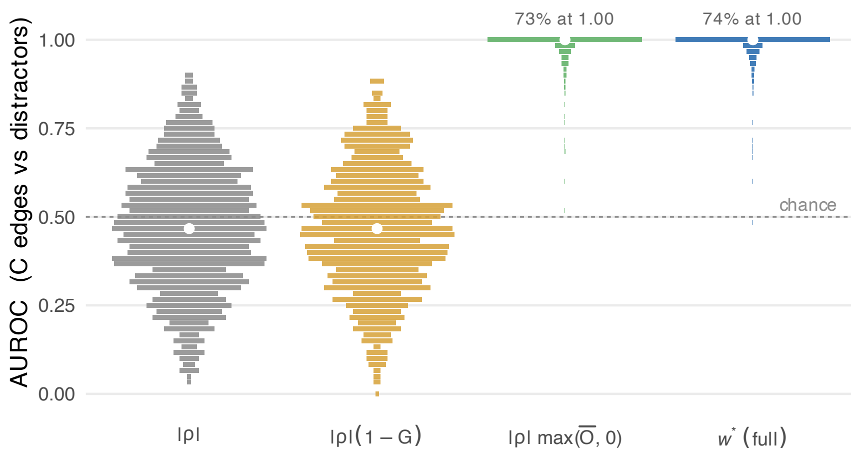

Figure 4: Ablation, estimator-level AUROC (within-C edges versus distractors) over 1,000 replicates; axis labels give each arm's scoring formula. AUROC over the 10×6 C–distractor pairs is discrete (grid 1/60), so each horizontal bar marks one attained value, with bar width proportional to the number of replicates at that value (normalized within arm); white dots mark medians. The concentration-only ablation $|\rho|(1-G)$ is at chance (mean .464); the coherence-only ablation $|\rho|\max(\bar{O}, 0)$ carries the effect (.985); the full w^* matches it (.985) — the a-priori prediction of Section 2.3. The right-hand distributions collapse onto a single dominant bar at ceiling (73–74% of replicates at AUROC exactly 1.00, annotated) with a short tail below. The oracle subgroup comparator (mean AUROC .002, inverted) is reported in the text as a diagnostic.

The end-to-end level tells a different and equally preregistered story. At $n = 600$ and $\pi_C = .30$ the raw pipeline already recovers the partition well, leaving little headroom: the per-replicate ARI distributions of the two pipelines nearly coincide (mean .946 raw versus .949 CW), and exact-C recovery is likewise indistinguishable (.815 versus .817; paired difference +.0020, CI spanning zero as reported above), and recovery of the correct number of communities was near ceiling in both arms (.987 raw, .973 CW). These results provide no evidence of an end-to-end improvement. Exact-C recovery depended strongly on the availability of true within-C edges in the shared raw-$|\rho|$ TMFG. When the graph retained the structural maximum of nine of the ten possible within-C edges, exact recovery was nearly certain (.99); below that maximum, it fell to .37 and was essentially the same for raw and CW. Because both arms used the same candidate graph, CW could reweight retained edges but could not restore edges removed by TMFG. The preregistered comparison therefore tests carrier-aware reweighting after raw- correlation edge selection, not carrier-aware edge selection itself.

The preregistered secondary analyses — the 36-condition parameter grid (generalization analyses) and bootstrap stability outcomes — have not yet been completed and are not reported here. The present report is limited to the primary condition; those secondary results will be reported separately.

4 Empirical illustration

The estimator originated in a concrete measurement problem: the Gray–Wheelwright Jungian Type Survey, whose thinking–feeling axis is the least well established of its three — coefficient alpha .38, against .72 for introversion–extraversion and .70 for intuition–sensation (Davis and Mattoon, 2006; Mattoon and Davis, 1995). This is the situation the estimator targets. If part of an axis is carried coherently by a minority of respondents, its pooled correlations are diluted toward the magnitude of incidental associations, and a candidate graph built on pooled correlation alone cannot tell the two apart (Section 2.3).

In the companion application that motivated this paper ($n = 605$ respondents to the 81-item Korean form, tetrachoric correlations, TMFG candidate graph, walktrap community detection), Eq. (3) replaced the raw edge weights inside an otherwise unchanged bootEGA pipeline. Its role there is exactly the estimator-level role validated in Section 3: an item-pair association that appears in the group average but is not carried by the respondents who carry its neighbouring edges is down-weighted before community detection and bootstrap stability assessment, so that item-retention decisions rest on coherently carried structure rather than on aggregation artefacts. Substantive results on the instrument — item selection, the refined forms, and their reliability — are reported in a separate manuscript (in submission) and are not claimed here; the illustration shows that the estimator's target phenomenon arises in real psychometric data and that the estimator deploys there without any modification of the surrounding pipeline.

5 Discussion

What the simulation establishes is deliberately narrow and, within that scope, unambiguous. At the estimator level, carrier coherence separates minority-carried structure from matched incoherent distractors at AUROC .985, with 73–74% of replicates at exactly 1.00; the mechanism operates exactly as the wedge argument predicts, and the separation is attributable to the coherence factor alone: the concentration-only ablation sits at chance. What the simulation does *not* establish is an end-to-end recovery advantage at $n = 600$ — Rule 2 failed, and the paper claims edge-level separation, not end-to-end recovery, for that reason. Rule 2 compared pipelines that shared one candidate graph, selected by raw correlation; it therefore tested reweighting, not selection.

The ablation result also disciplines the formula itself. The coherence-only ablation performed indistinguishably from the full estimator. Because the formula was locked, this licenses discussion of the $(1 - G)$ factor's role — not its post-hoc revision — and the honest reading is sharper than "neither helps nor harms." Analytically, G depends only on the magnitudes $|S_{re}|$, never on their signs, and is therefore direction-blind: toy scenarios A and B have identical G by construction, so the factor cannot contribute to detection. Concentration of support is, moreover, the mathematical signature of *any* minority-carried edge, coherent or spurious, and the concentration information that matters in the network context is already accounted for by $\bar{O}$, conditionally on alignment: misaligned concentration depresses the carrier cosines through their normalization, while aligned concentration is precisely the signal. What $(1 - G)$ adds beyond $\bar{O}$ is an *unconditional* penalty on minority-ness itself, in direct tension with the estimator's motivation. A deterministic dilution sequence makes the tension concrete (Table 1). It is computed exactly by hand from the toy construction, hence not tuned on confirmatory data: the three-carrier scenario-A structure is held fixed while the all-zero background grows. Both evidence terms ($|\rho|$ and $\bar{O}$) strengthen monotonically, yet the full weight w^* is non-monotone and lowest exactly where the evidence is strongest, with $w^* \to 0$ as the coherent minority becomes rarer — the paper's own headline use-case. Nor does $(1 - G)$ serve as an uncertainty proxy: it depends on the carried *fraction* of the sample, whereas sampling precision scales with the absolute *number* of carrying respondents, an inferential question that belongs to resampling downstream, not to the point estimator. Equation (3) retains $(1-G)$ because the preregistration locked it; retention is preregistration discipline, not endorsement. Whether a Gini-free variant, or one weighted by the effective number of carriers rather than their fraction, should replace it is the subject of the next preregistration; concentration would in any case remain reported per edge, as an annotation giving carrier count and carrier share alongside the weight. The preregistration's own interpretation table anticipated exactly this contingency.

Table 1: Deterministic dilution of the scenario-A triangle: the three carriers are held fixed while the all-zero background grows. Every value is an exact hand computation from the toy construction (no simulation, no tuning). Both evidence terms strengthen monotonically; the $(1 - G)$ factor drives the full CW weight down exactly where the evidence is strongest, while the Gini-free product $|\rho| \max(\bar{O}, 0)$ is monotone in the evidence.

carriers	ρ	$\bar{O}$	$1 - G$	w^* (CW)	$\lvert\rho\rvert \max(\bar{O}, 0)$
3/18 (16.7%)	.397	+.667	.712	.188	.265
3/36 (8.3%)	.651	+.947	.379	.234	.617
3/60 (5.0%)	.737	+.985	.234	.170	.726
3/120 (2.5%)	.804	+.997	.121	.097	.802

Three limitations bound the claims. First, when two carrier subgroups couple the same item pair with opposite signs in equal measure, the pooled correlation itself cancels toward zero, and no reweighting of a near-zero edge can rescue it; a coherence statistic that tracks positive and negative alignment separately is future work. Second, the method addresses between-respondent heterogeneity in cross-sectional data and says nothing about within-person dynamics. Third — and most consequentially — the end-to-end analysis shows that the current pipeline's bottleneck is upstream of the estimator: raw-correlation TMFG can discard true minority-carried edges that no subsequent reweighting can recover. Carrier-aware candidate-edge selection therefore requires a new, independently preregistered study. Companion work extending the carrier construction to directed, flow-like associations and to permutation-based inference is developed elsewhere.

Reproducibility statement

The preregistration and its pre-confirmatory amendment are archived, date-stamped, in the project repository alongside the engine (`sim1_engine.R`), the analysis and figure scripts, and an md5 manifest of all outputs. All seeds are published (calibration 20260718, discarded; confirmatory 202608250001–202608251000; control 202608260001–202608261000), and the deterministic map from nominal seed IDs to random-number-generator seeds is documented in the engine. The confirmatory run used R 4.5.1 with `psych` 2.6.5, `NetworkToolbox` 1.4.4, and `igraph`; figures were produced by a single script from the archived outputs. The repository is public at https://github.com/ad0c1/CW_unified.

References

Christensen, A. P., and Golino, H. (2021). Estimating the stability of psychological dimensions via bootstrap exploratory graph analysis: A Monte Carlo simulation and tutorial. *Psych*, 3(3), 479–500.

Davis, M. F., and Mattoon, M. A. (2006). Reliability and validity of the Gray-Wheelwrights Jungian Type Survey. *European Journal of Psychological Assessment*, 22(4), 233–239.

Epskamp, S., and Fried, E. I. (2018). A tutorial on regularized partial correlation networks. *Psychological Methods*, 23(4), 617–634.

Fisher, A. J., Medaglia, J. D., and Jeronimus, B. F. (2018). Lack of group-to-individual generalizability is a threat to human subjects research. *Proceedings of the National Academy of Sciences*, 115(27), E6106–E6115.

Golino, H. F., and Epskamp, S. (2017). Exploratory graph analysis: A new approach for estimating the number of dimensions in psychological research. *PLoS ONE*, 12(6), e0174035.

Jamison, L., Christensen, A. P., and Golino, H. F. (2024). Metric invariance in exploratory graph analysis via permutation testing. *Methodology*, 20(2), e12877.

Massara, G. P., Di Matteo, T., and Aste, T. (2017). Network filtering for big data: Triangulated maximally filtered graph. *Journal of Complex Networks*, 5(2), 161–178.

Mattoon, M. A., and Davis, M. (1995). The Gray-Wheelwrights Jungian Type Survey: Development and history. *Journal of Analytical Psychology*, 40(2), 205–234.

Molenaar, P. C. M. (2004). A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever. *Measurement: Interdisciplinary Research and Perspectives*, 2(4), 201–218.

Pons, P., and Latapy, M. (2006). Computing communities in large networks using random walks. *Journal of Graph Algorithms and Applications*, 10(2), 191–218.

A Preregistration deviations

One amendment, made before any confirmatory replicate was run and authorized by the preregistration's revision rule ("any revision creates Simulation 1b with a new file, seeds, and decision rules"):

Parameter	Preregistered	Amended (1b)	Reason
Pair factor $D_{r,d}$	$N(0,1)$	$N(1,1)$	zero-mean factor saturates under dichotomization (concordant-1 cell capped at $\pi_D/2$)
Distractor prevalence π_D	.10	.15	calibration target unreachable at .10 for any δ (δ-grid evidence archived in amendment file)
Confirmatory seed block	2026071800xx	2026082500xx	original block retired unused

The calibration target, decision rules, and all other parameters are unchanged. The original seed block was never executed; the amendment file records the full δ-grid from the discarded pilot.